

## ORIGINAL RESEARCH

## Association Between Size of Left Ventricle and Long-Term Cardiovascular Events

Yun-Jiu Cheng , MD, PhD\*; Zi-Li Liao, MD, PhD\*; Li-Juan Liu , MD, PhD; Yue-Dong Ma , MD, PhD\*; Xu-Miao Chen, MD, PhD\*; Ze-Xuan Wu , MD, PhD; Hui-Qiang Wei , MD, PhD; Wei-Dong Lin , MD, PhD; Yi-Jian Liao, MD; Su-Hua Wu , PhD; Yu-Mei Xue , MD, PhD; Li-Chun Wang , MD, PhD; Yang Wu , MD, PhD

**BACKGROUND:** The prognostic significance of **small left ventricle (LV)** in the general population and its association with cardiovascular disease remain elusive.

**METHODS:** Using data from the UK Biobank, 39363 participants underwent baseline cardiovascular magnetic resonance imaging and were categorized into **3 groups based on left ventricular end-diastolic volume index: small, normal, and large LV**. Cox proportional hazards models were applied to evaluate the association between LV size and composite cardiovascular outcomes.

**RESULTS:** An elevated prevalence of **small LV** was observed among the 4305 participants, with a notable trend toward increased occurrence in older individuals. Over a median follow-up of 5.21 years, 1500 cases of major adverse cardiovascular events, 1096 coronary heart disease, 288 ischemic stroke, and 722 deaths from any cause occurred. After adjusting for multiple confounders, hazard ratios (HRs) of small LV comparing the normal LV were 1.24 (95% CI, 1.07–1.44) for major adverse cardiovascular events, 1.29 (95% CI, 1.09–1.53) for coronary heart disease, 1.53 (95% CI, 1.11–2.09) for ischemic stroke, and 1.33 (95% CI, 1.08–1.64) for death from all causes. Notably, a **U-shaped relationship was identified between left ventricular end-diastolic volume index and cardiovascular outcomes**. **Subgroup analysis** revealed that, compared with women, the presence of a small LV in men exhibited a significant correlation with adverse cardiovascular events, thereby demonstrating potential value in disease prediction and risk stratification.

**CONCLUSIONS:** These findings highlight the importance of considering small LV as a prognostic marker for cardiovascular disease and underscore the need for further research to elucidate the underlying mechanisms and develop targeted interventions.

**Key Words:** cardiac magnetic resonance imaging ■ cardiovascular disease ■ left ventricular end-diastolic volume index ■ small left ventricle

Cardiovascular disease (CVD) stands as the leading global contributor to morbidity and mortality.<sup>1</sup> Over the past few decades, numerous risk factors underlying CVD outcomes have been identified and elucidated, significantly advancing our understanding of cardiovascular mechanisms and facilitating the exploration of novel therapeutic approaches.<sup>2–5</sup> Over the

past few decades, a multitude of risk factors underlying CVD outcomes have been elucidated. Among these, left ventricular size emerges as a pivotal prognostic marker in cardiac pathologies. The dilated left ventricle (LV) is universally acknowledged as a harbinger of poor outcomes in numerous cardiac disorders.<sup>6–8</sup> Conversely, the existence and potential consequences

Correspondence to: Yang Wu, Li-Chun Wang, Department of Cardiology, the First Affiliated Hospital, Sun Yat-Sen University, Guangzhou, and NHC Key Laboratory of Assisted Circulation Sun Yat-Sen University, Guangzhou, China. Email: [wuyang39@mail2.sysu.edu.cn](mailto:wuyang39@mail2.sysu.edu.cn); [wanglich@mail.sysu.edu.cn](mailto:wanglich@mail.sysu.edu.cn) and Yu-Mei Xue, Guangdong Provincial Geriatrics Institute, Guangdong Provincial People's Hospital (Guangdong Academy of Medical Sciences), Southern Medical University, Guangzhou, Guangdong, China. Email: [xyngdci@163.com](mailto:xyngdci@163.com)

\*Y.-J. Cheng, Z.-L. Liao, L.-J. Liu, Y.-D. Ma, and X.-M. Chen contributed equally.

This article was sent to Tazeen H. Jafar, MD, MPH, Associate Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/JAHA.124.040490>

For Sources of Funding and Disclosures, see page 13.

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## CLINICAL PERSPECTIVE

### What Is New?

- Small left ventricle size, measured by cardiac magnetic resonance, independently predicts adverse cardiovascular outcomes in the general population, with a U-shaped risk relationship between left ventricle size and events.

### What Are the Clinical Implications?

- Incorporating cardiac magnetic resonance-based left ventricle size assessment into risk stratification, especially for men, may improve early identification and targeted interventions to reduce cardiovascular morbidity and mortality.

## Nonstandard Abbreviations and Acronyms

|               |                                             |
|---------------|---------------------------------------------|
| <b>LVEDVI</b> | left ventricular end-diastolic volume index |
| <b>MACE</b>   | major adverse cardiovascular events         |
| <b>TAVR</b>   | Transcatheter aortic valve replacement      |

of a smaller LV have been largely neglected, revealing a skewed research emphasis and a significant knowledge gap in this area.

Despite previous studies investigating the relationship between small LV size and cardiovascular risks in populations with atrial fibrillation and cohorts post transcatheter aortic valve replacement, the association between small LV size and cardiovascular risks in the general population remains elusive.<sup>9,10</sup> Furthermore, these studies predominantly relied on echocardiography to define small LV size, which is known to have limitations in accuracy. Cardiac magnetic resonance imaging (CMR) is the gold standard for assessing cardiac structure and function, underscoring the inadequacy of echocardiography in this context.<sup>11</sup> Additionally, the lack of adjustment for body surface area in prior research calls into question the precision of their findings. Currently, there is a paucity of epidemiological data and insights into the CVD risk associated with CMR-detected small LV size in the general population.

Given these gaps, leveraging the vast UK Biobank cohort, our study aims to (1) determine the prevalence and risk factors of small LV size in the general population; (2) assess the cardiovascular outcomes associated with small LV size in the general population; and (3) explore, through subgroup analyses, whether cardiovascular risk factors modify the relationship between small LV size and cardiovascular prognosis.

## METHODS

### Availability of Data and Materials

The data that support the findings of this study are available from the UK Biobank project site, subject to registration and application processes. Further details are available at <https://www.ukbiobank.ac.uk>.

### Ethical Approval and Consent to Participate

The UK Biobank study obtained informed consent from the study participants and approval from its institutional review board. Scientific approval for this study was granted by the UK Biobank (application number:121022).

### Study Design and Participant Cohort

The UK Biobank, established in 2006, has accomplished a remarkable milestone by enrolling an extensive cohort exceeding half a million participants residing in communities, spanning ages 37 to 73. This comprehensive enrollment was accomplished through collaboration with 22 centers dispersed throughout the United Kingdom.<sup>12</sup> The baseline assessment within the UK Biobank study entailed comprehensive profiling of participant demographics, lifestyle habits, clinical history, cardiovascular medications, laboratory findings, and CMR findings. Participants unable to undergo this due to discomfort or ill health were excluded from recruitment. The protocol governing the UK Biobank is publicly accessible, ensuring transparency (<https://www.ukbiobank.ac.uk/wp-content/uploads/2011/11/uk-biobank-protocol.pdf>). Expanding its scope in 2015, the UK Biobank imaging initiative targets a randomized subset of 100 000 original participants for imaging, notably featuring detailed CMR imaging among other modalities.<sup>13</sup> This study enrolled 39 607 consecutive UK Biobank participants. Participants with incomplete CMR data, biologically implausible left ventricular end-diastolic volume index (LVEDVI) values (eg, <30 mL/m or >200 mL/m), or missing covariates were excluded to ensure data integrity. Extreme values were identified using a boxplot method (interquartile range  $\pm 1.5$ ) and validated by clinical experts. This study analyzed CMR examinations from 39 363 consecutive UK Biobank participants, contributing to a robust data set for medical research.

### CMR Image Acquisition

CMR examinations were conducted adhering to a predefined acquisition protocol using 1.5 Tesla scanners (MAGNETOM Aera, Siemens Healthcare, Syngo Platform VD13A).<sup>14</sup> The protocol encompassed standard long-axis views and a comprehensive short-axis

stack spanning the entire ventricular length, acquired via balanced steady-state free precession sequences. A fully automated, quality-assured pipeline facilitated CMR image analysis.

In our comprehensive assessment, we evaluated pivotal indices pertaining to the structural and functional characteristics of the LV. These encompassed the left ventricular EDV, left ventricular end-systolic volume, and left ventricular stroke volume, which serve as fundamental metrics of cardiac performance. Additionally, we calculated the corresponding indices for each of these parameters, namely left ventricular EDVi, left ventricular end-systolic volume index, and left ventricular stroke volume index. These indices normalize the respective volumes by body surface area, providing a more accurate assessment of LV function across individuals of varying sizes. Lastly, the left ventricular ejection fraction (EF) was also measured, serving as a crucial indicator of LV systolic function and contractility. It is noteworthy that left ventricular EDVi has emerged as a widely accepted metric in cardiovascular medicine, offering a more precise assessment of left ventricular size.<sup>15–19</sup>

## LV Size Assessment

The LV size was quantified using left ventricular EDVi derived from CMR. In light of the nascent concept of a small LV, a universal definition remains elusive. Therefore, we adopted the criteria derived from a recent, comprehensive meta-analysis of CMR studies on cardiac structures, published in a prestigious journal and encompassing the largest sample size to date.<sup>20</sup> These normative reference ranges are widely accepted, spanning 60 to 109 mL/m<sup>2</sup> for men and 56 to 96 mL/m<sup>2</sup> for women. Consequently, a small LV was defined as an left ventricular EDVi <60 mL/m<sup>2</sup> in men or 56 mL/m<sup>2</sup> in women.

## Ascertainment of Outcomes

The primary study outcome was major adverse cardiovascular events (MACE), a composite of coronary heart disease (*International Classification of Diseases, Tenth Revision* [ICD-10]: I20–I25), heart failure (ICD-10: I50), stroke (ICD-10 of ischemic stroke: I63; ICD-10 of hemorrhagic stroke: I61), and cardiovascular death (ICD-10 of the cause for death: I00–I99). We examined coronary heart disease, heart failure, ischemic stroke, and death from any cause as secondary outcomes.

## Statistical Analysis

To characterize baseline attributes stratified by LV size categories (normal, small, and large), descriptive statistical methods were employed to calculate mean values with accompanying SDs, alongside frequencies and

percentages for categorical variables. The baseline characteristics across the 3 LV size categories were statistically compared using the Kruskal–Wallis test for continuous variables and the chi-square test for categorical variables. Univariate logistic regression analyses were employed to identify significant correlates between baseline variables and the presence of a small LV size. Multivariable logistic modeling was used to explore these relationships further, adjusting for age, sex, race, less education, never drinking, current smoking, clinical history, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index (BMI), and laboratory findings.

In our analysis, missing data were addressed through the application of full information maximum likelihood estimation, assuming data were missing at random. Before their incorporation into the fully adjusted models, any missing values pertaining to sociodemographic covariates were imputed using the Markov Chain Monte Carlo multiple imputation method. Cox proportional hazard regression was employed to estimate the associations of 3 LV size categories and MACE, death from any cause, and noncardiovascular death. The results are reported as hazard ratios (HRs) and 95% CIs. Model 1 was adjusted for age, sex, and race; Model 2 was adjusted by Model 1 variables plus education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, BMI, and total cholesterol. Our fully adjusted model, Model 3, was adjusted by Model 2 variables plus hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and left ventricular EF. The cumulative incidence rates of the aforementioned outcomes, occurring subsequent to the visit, were quantified per 1000 person-years. *P* values for all outcomes were derived from log-rank tests, ensuring statistical rigor. Cumulative incidence curves depicting time to first event, stratified by LV categories, were constructed using the Kaplan–Meier method and compared using the standard log-rank test for multiple groups. Additionally, the hazard of cardiovascular outcomes was estimated using Cox proportional hazard analysis, with the analysis stratified according to quintiles of left ventricular EDVi.

The nonlinear relationship between baseline left ventricular EDVi and outcomes was assessed via multivariable Cox models employing restricted cubic splines. To ascertain the consistency of this nonlinear association across predefined subgroups, we refined our analysis by incorporating adjusted restricted cubic splines and evaluated cross-subgroup interactions within strata of age, sex, hypertension, diabetes, current drinking, current smoking, BMI, and left ventricular EF.

A series of rigorous sensitivity analyses were undertaken to confirm the robustness of our findings.

Initially, Cox proportional hazards regression models were applied to evaluate the impact of heart failure, atrial fibrillation, hemorrhagic stroke, cardiovascular death, and noncardiovascular death. Subsequently, the relationship between LV size and cardiovascular outcomes was reassessed, employing UK Biobank-specific cutoff values to ensure robustness of the results.<sup>11</sup> Furthermore, the Fine–Gray model was leveraged to account for competing risks and assess the associations between LV size and cardiovascular outcomes. Lastly, a repeated Cox regression analysis was conducted, excluding individuals with baseline CVD, to further validate our conclusions.

Sensitivity analyses further validated the robustness of our findings. First, redefining LV size using UK Biobank-specific cutoffs yielded results nearly identical to our primary analysis, confirming that the associations are not artifacts of arbitrary LV size thresholds. Second, competing risk analysis accounting for noncardiovascular death demonstrated that the elevated risk of MACE in individuals with small LVs persists independently of competing mortality, excluding confounding by differential survival. Third, excluding participants with baseline CVD eliminated reverse causality concerns, showing that small LV predicts incident events rather than reflecting preexisting disease. Collectively, these analyses address 3 key limitations—definitional variability, competing risks, and reverse causation—and confirm that the U-shaped relationship between left ventricular EDVi and cardiovascular risk is consistent across methodological approaches.

To investigate the potential influence of confounding factors on MACE among patients with varying LV sizes, subgroup analyses were undertaken. These analyses were stratified by age (dichotomized at 65 years), hypertension status, diabetes, current drinking, current smoking, BMI (categorized as  $\geq 25$  or  $< 25$  kg/m<sup>2</sup>), and left ventricular EF (dichotomized at 50%). To refine our understanding of the relationships, we employed adjusted restricted cubic spline models and assessed cross-subgroup interactions within subgroups defined by the aforementioned factors. Furthermore, Cox proportional hazards regression analyses were conducted to quantify the associations between LV size and cardiovascular outcomes, with results stratified by sex.

Statistical analyses were conducted in Stata version 16.0 (StataCorp LP, College Station, TX) and R version 4.2.1 (R Foundation for Statistical Computing).

## RESULTS

### Baseline Characteristics

The exclusion criteria excluded 244 participants from the 39 607 participants. Finally, 39 363 participants undergoing baseline CMR were analyzed from the UK

Biobank. [Table 1](#) presents the baseline demographic and clinical characteristics of the study cohort, stratified by LV size categories: small LV (n=4305), normal LV (n=33 991), and large LV (n=1067). Individuals with a small LV exhibited a higher prevalence of advanced age, male sex, current smoking, elevated Townsend deprivation index, systolic and diastolic blood pressure, BMI, reduced physical activity levels, and a more frequent history of coronary heart disease, stroke, heart failure, hypertension, and diabetes compared with those with normal LV dimensions. Conversely, they were less likely to be male, White, current drinkers, or possess higher education levels. Laboratory findings revealed increased levels of hemoglobin concentration, white blood cell counts, C-reactive protein, fasting glucose, hemoglobin A1c, low-density lipoprotein cholesterol, and triglycerides among subjects with a small LV. CMR analysis specifically in this subgroup demonstrated a higher left ventricular EF coupled with reduced LV dimensions. In the context of heart failure, the prevalence peaked within the large LV group, attaining 27.89% for overall heart failure and 26.74% specifically for heart failure with reduced EF. Conversely, the small LV group exhibited notably lower rates of 3.27% for overall heart failure and 2.60% for heart failure with reduced EF ([Figure S1](#)).

### Prevalence and Predictors of Small LV in General Population

In [Figure S2](#), we present the prevalence of varying LV sizes across distinct age cohorts, ranging from 45 to 49 to  $\geq 75$  years. A notable trend emerges regarding the prevalence rate of small LV, which increases progressively with advancing age. Initially, the prevalence of large LV surpasses that of small LV, but as age progresses, the latter gains prominence. Specifically, the incidence of small LV commences at 2.7% in the 45 to 49 age group, escalating to 19.7% among individuals aged  $\geq 75$ , indicative of a pronounced age-related shift.

In a multivariable logistic regression analysis, independent correlates for the presence of a small LV size were identified as age, female sex, non-White race, never drinking, current smoking, heart failure, hypertension, diabetes, Townsend deprivation index, systolic blood pressure, diastolic blood pressure, BMI, hemoglobin, C-reactive protein, hemoglobin A1c, and triglycerides were independent correlates for the presence of small LV ([Table S1](#)).

### Association of Small LV With Cardiovascular Outcomes

Over a median follow-up period of 5.21 (interquartile range, 4.38–6.54) years, 1500 cases of MACE, 1096



**Table 1. Baseline Characteristics According to LV Size**

| Factor                                                    | Overall<br>(n=39363) | Normal LV<br>(n=33991) | Small LV<br>(n=4305) | Large LV<br>(n=1067) | P value |
|-----------------------------------------------------------|----------------------|------------------------|----------------------|----------------------|---------|
| Age, y                                                    | 63.6 (7.6)           | 63.3 (7.5)             | 66.9 (6.8)           | 62.5 (8.0)           | <0.001  |
| Men, %                                                    | 18836 (47.9)         | 16277 (47.9)           | 2003 (46.5)          | 556 (52.1)           | 0.005   |
| White race, %                                             | 38258 (97.2)         | 33120 (97.4)           | 4107 (95.4)          | 1031 (96.6)          | <0.001  |
| Higher education, %                                       | 18021 (45.8)         | 15756 (46.4)           | 1744 (40.5)          | 521 (48.8)           | <0.001  |
| Current drinking, %                                       | 37492 (95.2)         | 31962 (94.0)           | 3958 (91.9)          | 985 (92.3)           | <0.001  |
| Current smoking, %                                        | 2458 (6.2)           | 1816 (5.3)             | 293 (6.8)            | 126 (11.8)           | <0.001  |
| Clinical history, %                                       |                      |                        |                      |                      |         |
| Coronary heart disease                                    | 889 (2.3)            | 338 (1.0)              | 62 (1.4)             | 16 (1.5)             | <0.001  |
| Stroke                                                    | 230 (0.6)            | 608 (1.8)              | 128 (3.0)            | 246 (23.1)           | 0.009   |
| Heart failure                                             | 982 (2.5)            | 608 (1.8)              | 128 (3.0)            | 246 (23.1)           | <0.001  |
| Atrial fibrillation                                       | 1089 (2.8)           | 902 (2.7)              | 130 (3.0)            | 57 (5.3)             | <0.001  |
| Hypertension                                              | 7464 (19.0)          | 9534 (28.0)            | 1749 (40.6)          | 317 (29.7)           | <0.001  |
| Diabetes                                                  | 1005 (2.6)           | 1541 (4.5)             | 496 (11.5)           | 48 (4.5)             | <0.001  |
| Townsend deprivation index                                | -1.9 (2.7)           | -1.9 (2.7)             | -1.8 (2.8)           | -1.9 (2.7)           | <0.001  |
| Physical activity, metabolic equivalent task-min/week     | 2451.2 (2385.6)      | 2465.4 (2386.3)        | 2215.3 (2318.4)      | 2903.9 (2524.5)      | <0.001  |
| Systolic blood pressure, mmHg                             | 132.6 (21.6)         | 132.2 (21.5)           | 135.5 (21.9)         | 132.6 (23.5)         | <0.001  |
| Diastolic blood pressure, mmHg                            | 79.9 (12.4)          | 79.7 (12.3)            | 82.1 (12.5)          | 77.4 (13.5)          | <0.001  |
| Body mass index, kg/m <sup>2</sup>                        | 26.5 (4.2)           | 26.4 (4.1)             | 27.6 (4.4)           | 26.1 (4.2)           | <0.001  |
| Cardiovascular medications, %                             |                      |                        |                      |                      |         |
| Antihypertensive                                          | 8244 (20.9)          | 6754 (19.9)            | 1248 (29.0)          | 242 (22.7)           | <0.001  |
| Antiplatelet                                              | 3250 (8.3)           | 2698 (7.9)             | 442 (10.3)           | 110 (10.3)           | <0.001  |
| Lipid lowering                                            | 7559 (19.2)          | 6123 (18.0)            | 1253 (29.1)          | 183 (17.2)           | <0.001  |
| Beta blocker                                              | 1992 (5.1)           | 1682 (4.9)             | 217 (5.0)            | 93 (8.7)             | <0.001  |
| Diuretics                                                 | 1328 (3.4)           | 1046 (3.1)             | 241 (5.6)            | 41 (3.8)             | <0.001  |
| Laboratory findings                                       |                      |                        |                      |                      |         |
| Hemoglobin, g/dL                                          | 14.2 (1.2)           | 14.2 (1.2)             | 14.4 (1.2)           | 14.0 (1.2)           | <0.001  |
| White blood cell count, ×10 <sup>3</sup> /mm <sup>3</sup> | 6.6 (1.8)            | 6.5 (1.8)              | 6.9 (1.8)            | 6.3 (1.6)            | <0.001  |
| Albumin, g/dL                                             | 45.4 (2.5)           | 45.4 (2.5)             | 45.2 (2.5)           | 45.5 (2.6)           | <0.001  |
| C-reactive protein, g/dL                                  | 2.0 (3.5)            | 2.0 (3.5)              | 2.6 (3.8)            | 1.8 (4.0)            | <0.001  |
| Fasting glucose, mmol/L                                   | 5.0 (1.0)            | 5.0 (0.9)              | 5.1 (1.2)            | 4.9 (0.8)            | <0.001  |
| Hemoglobin A1c, %                                         | 35.0 (5.2)           | 34.8 (4.9)             | 36.6 (6.6)           | 34.5 (4.7)           | <0.001  |
| Total cholesterol, mmol/L                                 | 4.7 (0.9)            | 4.7 (0.9)              | 4.7 (1.0)            | 4.5 (0.9)            | <0.001  |
| Low-density lipoprotein cholesterol, mmol/L               | 3.6 (0.8)            | 3.6 (0.8)              | 3.7 (0.9)            | 3.4 (0.8)            | <0.001  |
| High-density lipoprotein cholesterol, mmol/L              | 1.3 (0.3)            | 1.4 (0.3)              | 1.3 (0.3)            | 1.4 (0.3)            | <0.001  |
| Triglycerides, mmol/L                                     | 1.6 (1.0)            | 1.6 (0.9)              | 2.0 (1.1)            | 1.4 (0.8)            | <0.001  |
| Cardiac magnetic resonance findings                       |                      |                        |                      |                      |         |
| Left ventricular EDV, mL                                  | 138.1 (35.9)         | 140.4 (28.5)           | 97.4 (16.0)          | 229.5 (73.7)         | <0.001  |
| Left ventricular EDV index, mL/m <sup>2</sup>             | 74.0 (16.2)          | 75.2 (11.0)            | 52.0 (5.1)           | 122.9 (36.8)         | <0.001  |
| Left ventricular ESV, mL                                  | 62.1 (23.2)          | 62.5 (16.9)            | 43.1 (10.6)          | 125.6 (69.5)         | <0.001  |
| Left ventricular ESV index, mL/m <sup>2</sup>             | 33.2 (11.3)          | 33.4 (7.3)             | 22.9 (4.4)           | 67.2 (36.4)          | <0.001  |
| Left ventricular SV, mL                                   | 76.0 (18.4)          | 77.9 (16.3)            | 54.3 (10.7)          | 103.9 (26.7)         | <0.001  |
| Left ventricular SV index, mL/m <sup>2</sup>              | 40.8 (8.4)           | 41.8 (6.9)             | 29.1 (4.6)           | 55.8 (12.8)          | <0.001  |
| Left ventricular ejection fraction, %                     | 55.5 (6.5)           | 55.7 (6.0)             | 55.9 (7.1)           | 47.5 (11.7)          | <0.001  |

Data are presented as mean±SD or No. (percentage) of participants.

EDV indicates end-diastolic volume; ESV, end-systolic volume; LV, left ventricle; and SV, stroke volume.

coronary heart disease, 392 heart failure, 288 ischemic stroke, and 722 deaths from any cause occurred. The incidence rates of small LV involvement in these outcomes were 10.2 (95% CI, 8.9–11.6 per 1000 person-years) for MACE, 7.6 (95% CI, 6.6–8.9 per 1000 person-years) for coronary heart disease, 2.0 (95% CI, 1.5–2.6 per 1000 person-years) for ischemic stroke, 1.5 (95% CI, 1.1–2.1 per 1000 person-years) for heart failure, and 4.5 (95% CI, 3.8–5.5 per 1000 person-years) for death from any cause.

Table 2 displays the unadjusted and multivariate-adjusted HRs for MACE, coronary heart disease, ischemic stroke, heart failure, and death from any cause, stratified by small and large LV dimensions, compared with normal LV. In the unadjusted analysis (Model 1), individuals with either small or large LVs exhibited significantly elevated risks for MACE, with HRs of 1.50 (95% CI, 1.30–1.74) and 1.85 (95% CI, 1.44–2.37), respectively. Notably, these associations persisted after accounting for multiple confounding variables in Models 2 and 3. Specifically, in the fully adjusted Model 3, the MACE HRs for small and large LV, compared with normal LV, were 1.24 (95%

CI, 1.07–1.44) and 1.86 (95% CI, 1.45–2.39), respectively, underscoring the robustness of the observed associations.

Similarly, the associations between small and large LV and coronary heart disease, ischemic stroke, and death from any cause mirrored those observed for MACE. In the fully adjusted Model 3, compared with individuals with normal LV, the HRs for coronary heart disease were 1.29 (95% CI, 1.09–1.53) for small LV and 1.42 (95% CI, 1.02–1.96) for large LV. For ischemic stroke, the corresponding HRs were 1.53 (95% CI, 1.11–2.09) for small LV and 1.82 (95% CI, 1.06–3.13) for large LV. Heart failure increases related risks only in large LV, not in small LV. Lastly, the HRs for death from any cause were 1.33 (95% CI, 1.08–1.64) for small LV and 1.35 (95% CI, 1.09–1.65) for large LV, respectively.

Throughout the study period, cumulative incidence functions consistently demonstrated a monotonic association between a small LV size and cardiovascular outcomes, including MACE, coronary heart disease, ischemic stroke, and death from any cause (Figure 1A and Figure 2). Although the unadjusted Kaplan–Meier

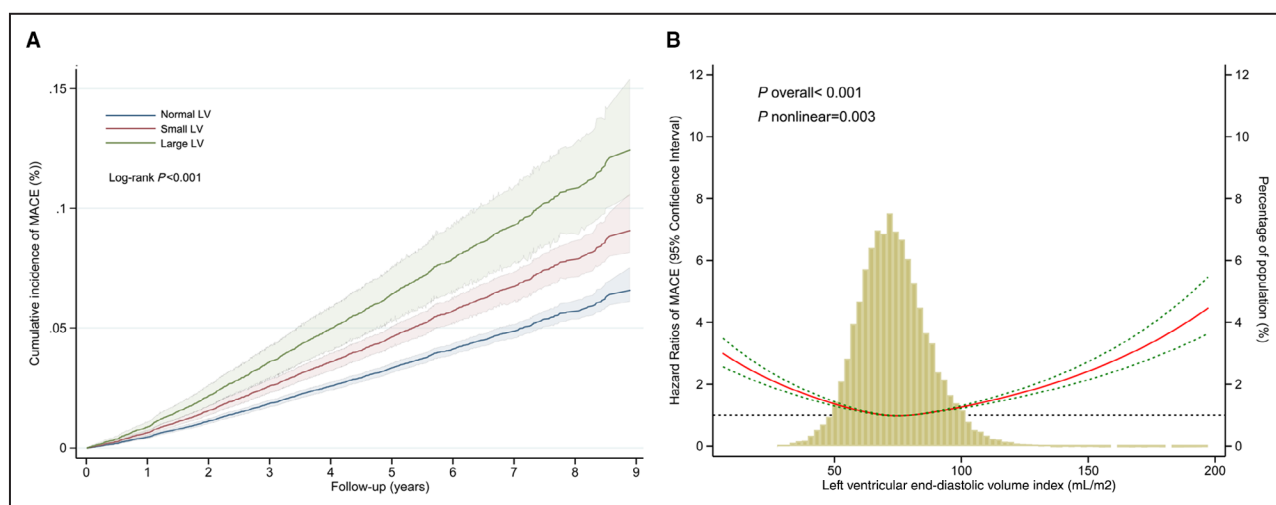
**Table 2. Association Between LV Size and Cardiovascular Outcomes**

| Outcomes                            | Events/person-y | Incidence rate (95% CI), per 1000 person-y | Risk difference (95% CI), per 1000 person-y | Model 1          | Model 2          | Model 3          |
|-------------------------------------|-----------------|--------------------------------------------|---------------------------------------------|------------------|------------------|------------------|
| Major adverse cardiovascular events |                 |                                            |                                             |                  |                  |                  |
| Normal LV                           | 1221/175800     | 6.9 (6.6–7.3)                              | Ref.                                        | Ref.             | Ref.             | Ref.             |
| Small LV                            | 213/20949       | 10.2 (8.9–11.6)                            | 3.2 (1.8–4.6)                               | 1.50 (1.30–1.74) | 1.29 (1.11–1.49) | 1.24 (1.07–1.44) |
| Large LV                            | 66/5080         | 13.0 (10.2–16.5)                           | 6.0 (2.9–9.2)                               | 1.85 (1.44–2.37) | 1.82 (1.42–2.34) | 1.86 (1.45–2.39) |
| Coronary heart disease              |                 |                                            |                                             |                  |                  |                  |
| Normal LV                           | 893/179172      | 5.0 (4.7–5.3)                              | Ref.                                        | Ref.             | Ref.             | Ref.             |
| Small LV                            | 165/21625       | 7.6 (6.6–8.9)                              | 2.6 (1.4–3.9)                               | 1.56 (1.32–1.85) | 1.33 (1.13–1.58) | 1.29 (1.09–1.53) |
| Large LV                            | 38/5294         | 7.2 (5.2–9.9)                              | 2.2 (0–4.5)                                 | 1.41 (1.02–1.95) | 1.39 (1.01–1.93) | 1.42 (1.02–1.96) |
| Ischemic stroke                     |                 |                                            |                                             |                  |                  |                  |
| Normal LV                           | 225/201613      | 1.1 (1.0–1.3)                              | Ref.                                        | Ref.             | Ref.             | Ref.             |
| Small LV                            | 49/24856        | 2.0 (1.5–2.6)                              | 0.9 (0.3–1.4)                               | 1.79 (1.31–2.44) | 1.57 (1.15–2.15) | 1.53 (1.11–2.09) |
| Large LV                            | 14/6338         | 2.2 (1.3–3.7)                              | 1.1 (0–2.3)                                 | 1.95 (1.14–3.35) | 1.79 (1.04–3.08) | 1.82 (1.06–3.13) |
| Heart failure                       |                 |                                            |                                             |                  |                  |                  |
| Normal LV                           | 293/201303      | 1.5 (1.3–1.6)                              | Ref.                                        | Ref.             | Ref.             | Ref.             |
| Small LV                            | 37/24871        | 1.5 (1.1–2.1)                              | 0 (–0.5–0.5)                                | 1.05 (0.74–1.47) | 0.83 (0.59–1.18) | 0.80 (0.56–1.13) |
| Large LV                            | 62/6039         | 10.3 (8.0–13.2)                            | 8.8 (6.1–11.4)                              | 6.96 (5.29–9.15) | 6.60 (5.01–8.71) | 6.85 (5.19–9.04) |
| Death from any cause                |                 |                                            |                                             |                  |                  |                  |
| Normal LV                           | 586/193526      | 3.0 (2.8–3.3)                              | Ref.                                        | Ref.             | Ref.             | Ref.             |
| Small LV                            | 109/23961       | 4.5 (3.8–5.5)                              | 1.5 (0.6–2.4)                               | 1.56 (1.27–1.91) | 1.38 (1.12–1.70) | 1.33 (1.08–1.64) |
| Large LV                            | 27/6114         | 4.4 (3.0–6.4)                              | 1.4 (0–3.1)                                 | 1.41 (0.96–2.08) | 1.31 (0.89–1.93) | 1.35 (1.09–1.65) |

Model 1: Adjusted for age, sex, and race.

Model 2: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, and total cholesterol.

Model 3: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, atrial fibrillation, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and left ventricular ejection fraction. LV indicates left ventricle.



**Figure 1. Associations between LV size and MACE.**

The cumulative incidence of MACE in subjects stratified by LV size (A) and adjusted associations of measures of left ventricular EDVi and MACE (B). The solid lines represent the adjusted HRs of left ventricular EDVi across the whole range and the dotted lines represent the corresponding 95% CIs. EDVi indicates end-diastolic volume index; HR, hazard ratio; LV, left ventricle; and MACE, major adverse cardiovascular events.

curves (Figures 1A and 2) demonstrate separation between LV size groups, the multivariable Cox models account for potential confounding factors, underscoring the independent prognostic value of LV size. This trend underscores the persistent link between reduced LV dimensions and increased risk for cardiovascular outcomes.

In Table 3, the relationship between quintiles of left ventricular EDVi and cardiovascular outcomes was presented, highlighting a notable trend. Specifically, compared with Quintile 3, both Quintile 1 (representing the lowest left ventricular EDVi) and Quintile 5 (the highest left ventricular EDVi) were associated with an elevated risk across various end points. However, it should be noted that Quintile 1 was not associated with an increased risk of heart failure, whereas Quintile 5 was. Figure 1B presented the associations between LV size and MACE, highlighting key trends. Figure 3, on the other hand, presented the adjusted association of left ventricular EDVi and cardiovascular outcomes. Specifically, it showed the adjusted associations of measures of left ventricular EDVi with coronary heart disease (Figure 3A), heart failure (Figure 3B, noting that the risk increases only in large LV, not in small LV), ischemic stroke (Figure 3C), and death from any cause (Figure 3D).

## Sensitivity Analyses

In the sensitivity analyses conducted to robustly evaluate the study findings, several notable observations were made. First, as presented in Table S2, small LV size did not significantly alter the risk of heart failure,

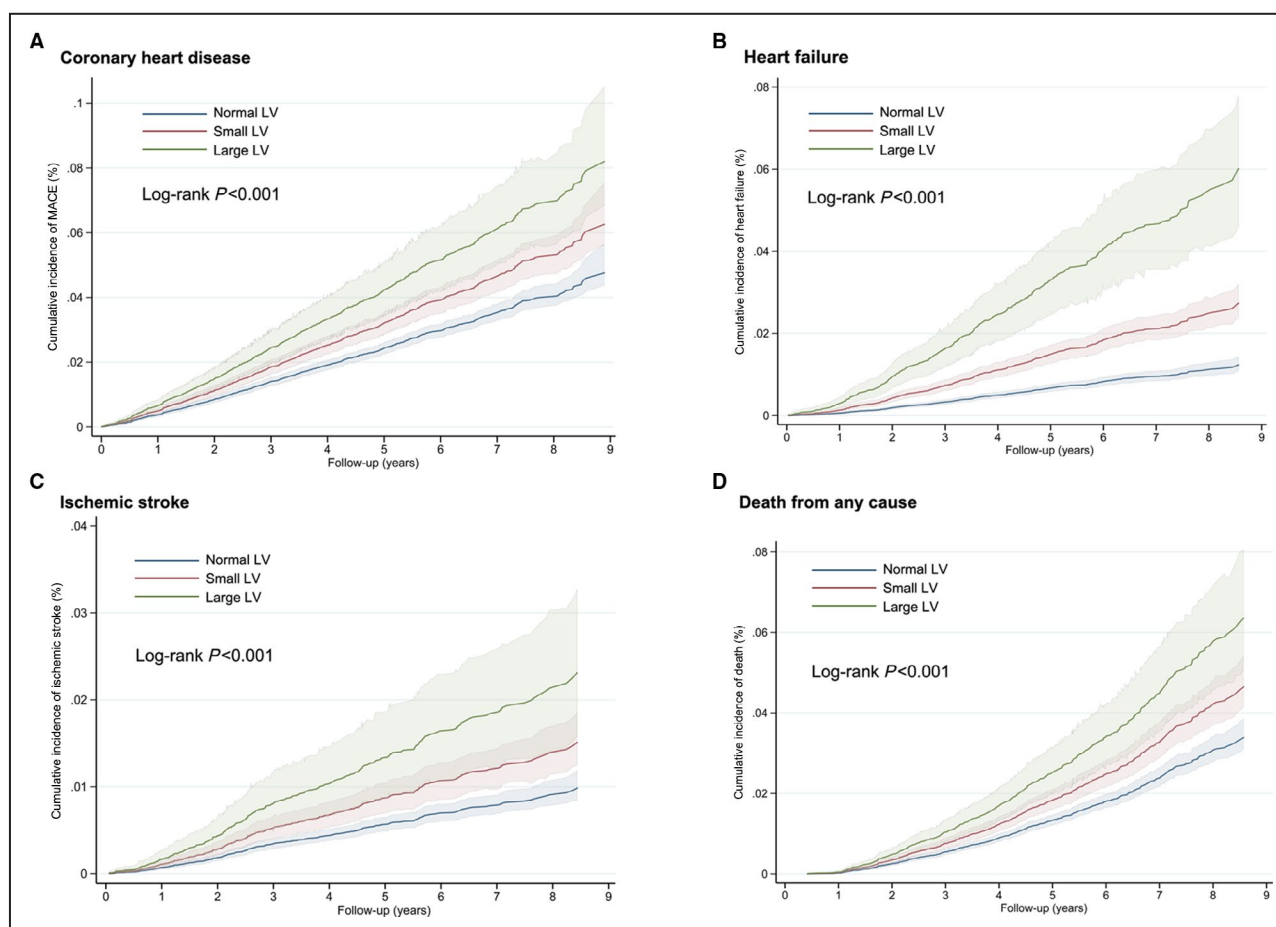
atrial fibrillation, hemorrhagic stroke, or cardiovascular death but was associated with an increased risk of noncardiovascular death (adjusted HR, 1.47 [95% CI, 1.17–1.84]). Conversely, large LV was found to elevate the risk of cardiovascular end points, including those mentioned previously, without significantly affecting the risk of noncardiovascular death.

Second, to assess the potential influence of LV size definitions, an additional analysis was performed using UK Biobank cutoff values to categorize small and large LVs. As reported in Table S3, this adjustment in definition did not appreciably alter the study results, suggesting the robustness of our findings against varying definitions of LV size.

Third, to account for the competing risk of death from any cause, a sensitivity analysis was conducted. The inclusion of death from any cause as a competing risk factor, as presented in Table S4, did not significantly affect the overall study conclusions, reinforcing the stability of our findings in the presence of this potential confounder.

Lastly, an analysis excluding subjects with baseline CVD was performed to evaluate the potential influence of preexisting conditions on the study outcomes. As detailed in Table S5, the exclusion of these subjects did not substantially alter the study results, further validating the robustness and generalizability of our findings.

Sensitivity analyses uniformly supported the primary findings. Redefining LV size thresholds, accounting for competing mortality, and excluding baseline CVD all yielded results congruent with the main models, with HRs for small LV remaining elevated across end points



**Figure 2. Kaplan–Meier curves for cardiovascular outcomes stratified by LV size.**

The cumulative incidence of coronary heart disease (A), heart failure (B), ischemic stroke (C), and death from any cause (D) in subjects stratified by LV size. Unadjusted Kaplan–Meier curves for cardiovascular outcomes stratified by LV size. LV indicates left ventricle; and MACE, major adverse cardiovascular events.

(Tables S3–S5). This methodological triangulation underscores the robustness of the U-shaped left ventricular EDVi–risk relationship.

## Stratified Analysis

In Figure 4, subgroup analyses were presented, stratified by demographic factors and MACE risk factors. Notably, the observed relationship between LV size and MACE risk remained consistent across various subgroups, including age, hypertension, diabetes, current drinking, current smoking, BMI and left ventricular EF, suggesting no significant effect modification by these factors. A notable exception was observed for sex, where both large and small LV sizes exhibited a modified risk of MACE, highlighting the potential sex-specific implications in the association between LV size and MACE outcomes ( $P$  for interaction=0.03). Adjusted restricted cubic splines fitted for multivariable Cox proportional model indicated a U-shaped association between left ventricular EDVi and MACE

stratified by risk factors (Figure 5). The nonlinear associations between left ventricular EDVi and the risk of MACE were persistent among subgroups stratified by age, hypertension, diabetes, current drinking, current smoking, and BMI ( $P$  nonlinear <0.001).

The baseline characteristics presented in Table S6 demonstrate a balanced distribution between male and female groups across different LV sizes, with statistical differences observed concurrently in the majority of baseline parameters for both groups.

Given the heightened risk of MACE associated with small LV in men, we further comprehensively examined the risk of other cardiovascular events in both sexes with varying ventricular sizes. As shown in Table S7, within Model 3, men with small LV exhibited an increased risk of coronary heart disease (adjusted HR, 1.30 [95% CI, 1.03–1.64]), whereas those with large ventricular size had an elevated risk as well (adjusted HR, 1.79 [95% CI, 1.21–2.67]). Similarly, small LV was associated with a higher risk of ischemic stroke in men (adjusted HR, 1.62 [95% CI, 1.02–2.56]), and large ventricular size



**Table 3. HRs (95% CIs) of Cardiovascular Outcomes According to Quintiles of Left Ventricular EDVi**

| Outcomes                                                   | Quintile 1       | Quintile 2       | Quintile 3       | Quintile 4       | Quintile 5        |
|------------------------------------------------------------|------------------|------------------|------------------|------------------|-------------------|
| Range of left ventricular EDVi (median), mL/m <sup>2</sup> | 36.4–62.0 (56.8) | 62.0–69.3 (65.9) | 69.3–76.0 (72.5) | 76.0–84.5 (79.8) | 84.5–286.3 (91.9) |
| Subjects                                                   | 7872             | 7873             | 7873             | 7873             | 7872              |
| Major adverse cardiovascular events                        |                  |                  |                  |                  |                   |
| Events                                                     | 326              | 284              | 255              | 297              | 338               |
| Incidence rate (95% CI), per 1000 person-y                 | 8.3 (7.5–9.3)    | 7.0 (6.2–7.8)    | 6.2 (5.5–7.0)    | 7.3 (6.5–8.1)    | 8.5 (7.6–9.4)     |
| Model 1                                                    | 1.35 (1.14–1.59) | 1.12 (0.95–1.33) | Ref.             | 1.17 (0.99–1.38) | 1.36 (1.16–1.60)  |
| Model 2                                                    | 1.63 (1.38–1.93) | 1.31 (1.11–1.55) | Ref.             | 1.29 (1.09–1.52) | 1.20 (1.02–1.41)  |
| Model 3                                                    | 1.44 (1.21–1.70) | 1.24 (1.04–1.47) | Ref.             | 1.24 (1.05–1.47) | 1.23 (1.04–1.44)  |
| Coronary heart disease                                     |                  |                  |                  |                  |                   |
| Events                                                     | 247              | 218              | 182              | 223              | 226               |
| Incidence rate (95% CI), per 1000 person-y                 | 6.1 (5.4–6.9)    | 5.3 (4.6–6.0)    | 4.4 (3.8–5.0)    | 5.4 (4.8–6.2)    | 5.5 (4.8–6.2)     |
| Model 1                                                    | 1.41 (1.17–1.71) | 1.21 (0.99–1.47) | Ref.             | 1.24 (1.02–1.51) | 1.25 (1.03–1.52)  |
| Model 2                                                    | 1.76 (1.45–2.13) | 1.44 (1.18–1.75) | Ref.             | 1.08 (0.89–1.31) | 1.39 (1.14–1.69)  |
| Model 3                                                    | 1.53 (1.26–1.86) | 1.35 (1.11–1.65) | Ref.             | 1.11 (0.91–1.35) | 1.34 (1.10–1.62)  |
| Ischemic stroke                                            |                  |                  |                  |                  |                   |
| Events                                                     | 76               | 60               | 48               | 42               | 62                |
| Incidence rate (95% CI), per 1000 person-y                 | 1.7 (1.3–2.1)    | 1.3 (1.0–1.7)    | 1.0 (0.8–1.4)    | 0.9 (0.7–1.2)    | 1.3 (1.0–1.7)     |
| Model 1                                                    | 1.86 (1.28–2.71) | 1.44 (0.97–2.14) | Ref.             | 1.15 (0.76–1.74) | 1.47 (0.99–2.18)  |
| Model 2                                                    | 2.13 (1.45–3.12) | 1.61 (1.08–2.40) | Ref.             | 1.23 (0.81–1.87) | 1.33 (0.90–1.98)  |
| Model 3                                                    | 1.84 (1.25–2.71) | 1.51 (1.02–2.25) | Ref.             | 1.19 (0.78–1.80) | 1.36 (0.92–2.02)  |
| Heart failure                                              |                  |                  |                  |                  |                   |
| Events                                                     | 52               | 40               | 74               | 63               | 163               |
| Incidence rate (95% CI), per 1000 person-y                 | 1.1 (0.9–1.5)    | 0.9 (0.6–1.2)    | 1.6 (1.3–2.0)    | 1.3 (1.1–1.7)    | 3.5 (3.0–4.1)     |
| Model 1                                                    | 0.85 (0.59–1.22) | 0.64 (0.43–0.95) | Ref.             | 1.18 (0.84–1.64) | 2.61 (1.95–3.49)  |
| Model 2                                                    | 0.98 (0.67–1.42) | 0.72 (0.48–1.07) | Ref.             | 1.27 (0.91–1.78) | 2.35 (1.76–3.15)  |
| Model 3                                                    | 0.79 (0.54–1.15) | 0.65 (0.44–0.97) | Ref.             | 1.20 (0.86–1.68) | 2.43 (1.81–3.25)  |
| Death from any cause                                       |                  |                  |                  |                  |                   |
| Events                                                     | 167              | 159              | 103              | 124              | 169               |
| Incidence rate (95% CI), per 1000 person-y                 | 3.9 (3.3–4.5)    | 3.6 (3.1–4.2)    | 2.3 (1.9–2.8)    | 2.7 (2.3–3.3)    | 3.8 (3.3–4.4)     |
| Model 1                                                    | 1.69 (1.32–2.15) | 1.56 (1.22–2.00) | Ref.             | 1.21 (0.93–1.57) | 1.62 (1.27–2.07)  |
| Model 2                                                    | 2.03 (1.59–2.61) | 1.82 (1.42–2.34) | Ref.             | 1.33 (1.02–1.73) | 1.42 (1.11–1.82)  |
| Model 3                                                    | 1.80 (1.40–2.31) | 1.72 (1.34–2.21) | Ref.             | 1.29 (0.99–1.68) | 1.45 (1.13–1.85)  |

Model 1: Adjusted for age, sex, and race.

Model 2: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, and total cholesterol.

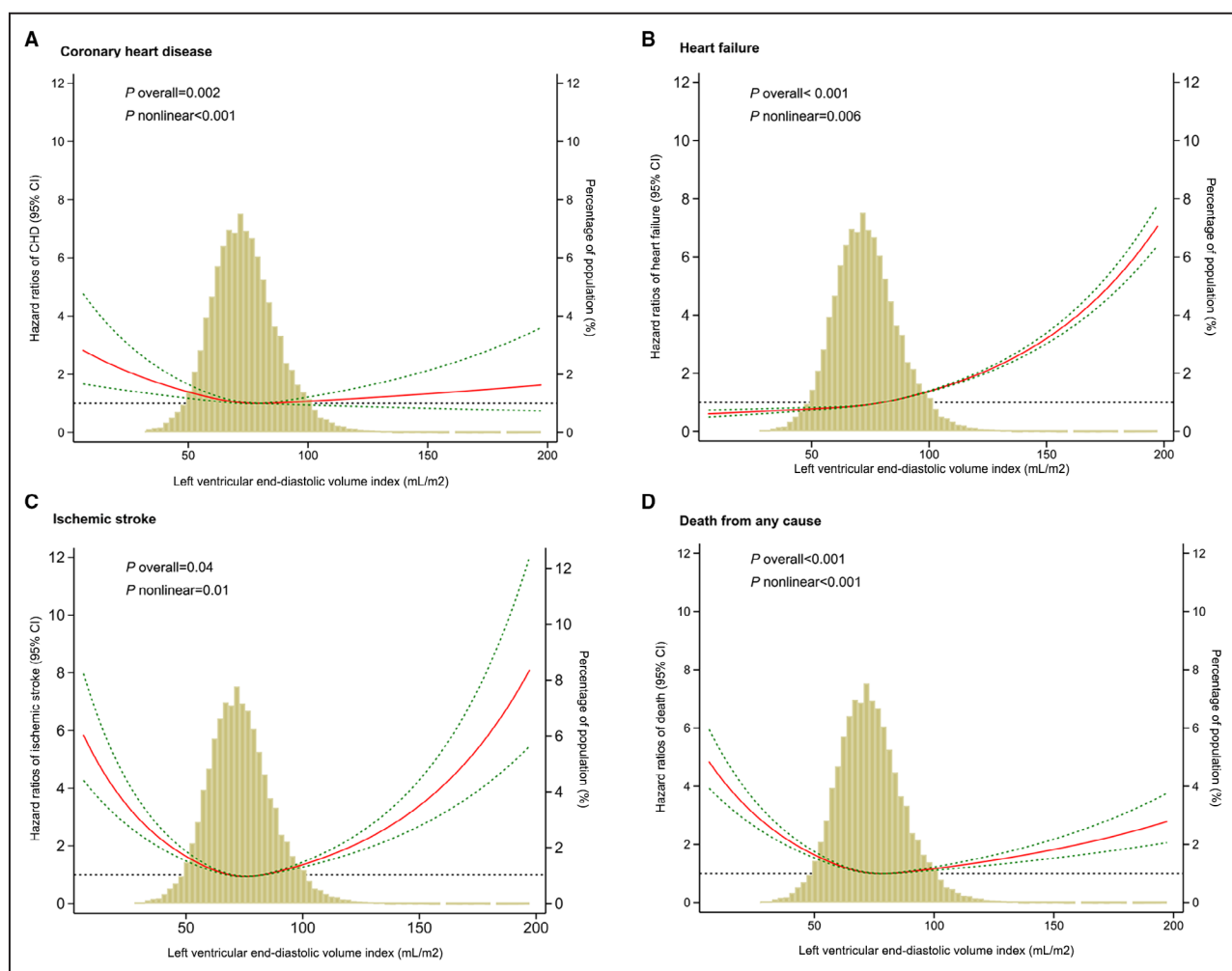
Model 3: Adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, atrial fibrillation, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and left ventricular ejection fraction. EDVi indicates end-diastolic volume index; and HR, hazard ratio.

further escalated this risk (adjusted HR, 1.58 [95% CI, 0.69–3.61]). Additionally, small LV increased the risk of death from any cause in men (adjusted HR, 1.36 [95% CI, 1.02–1.80]), with large LV also contributing to this risk (adjusted HR, 1.31 [95% CI, 0.78–2.21]). Notably, neither small nor large LV was found to significantly elevate the risk of these cardiovascular events in women.

## DISCUSSION

### Summary of Findings

Leveraging the extensive UK Biobank cohort, the present study presents compelling evidence for the correlation between reduced LV dimensions and detrimental cardiovascular outcomes in the general



**Figure 3. Adjusted association of left ventricular EDVi and cardiovascular outcomes.**

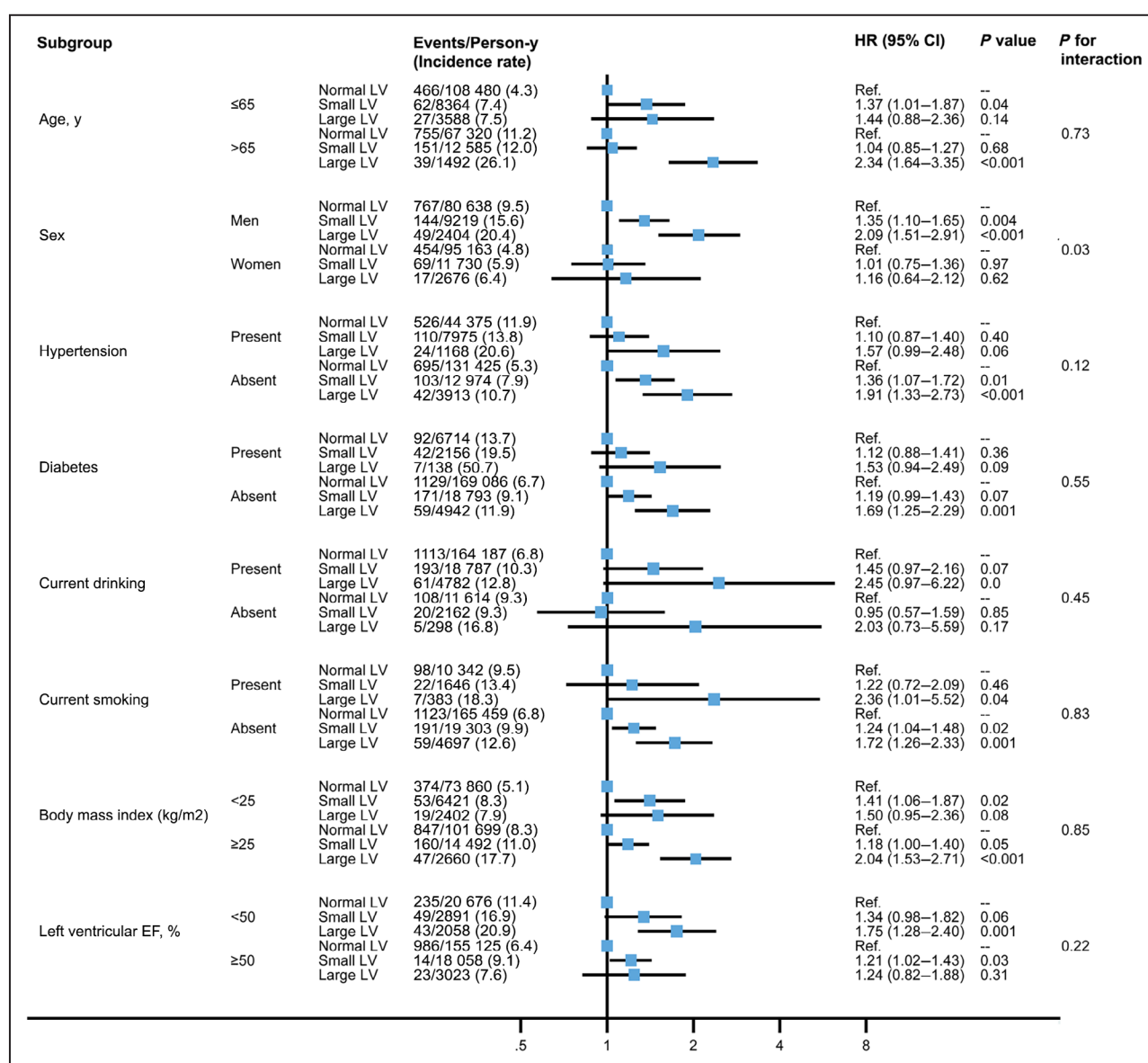
Adjusted associations of measures of left ventricular EDVi and coronary heart disease (A) heart failure (B) ischemic stroke (C) and death from any cause (D). The solid lines represent the adjusted HRs of left ventricular EDVi across the whole range and the dotted lines represent the corresponding 95% CIs. CHD indicates coronary heart disease; EDVi, end-diastolic volume index; and HR, hazard ratio.

populace. It identifies the prevalence and significant risk factors associated with small LV size in the general population, with age and various clinical and demographic factors emerging as key predictors. By meticulous examination of CMR imaging data, we established that a small LV size, operationally defined as left ventricular EDVi  $<60$  mL/m<sup>2</sup> for men and  $56$  mL/m<sup>2</sup> for women, stands as an independent predictor of MACE, coronary heart disease, ischemic stroke, death from any cause and noncardiovascular death. Notably, although small LV did not exacerbate the risk of adverse outcomes in women, it was significantly associated with heightened risks of MACE, coronary heart disease, ischemic stroke, and death from any cause in men. These findings underscore the significance of incorporating LV size as a potential prognostic indicator, particularly

given its previous underappreciation in the research landscape.

### Comparison With Current Studies

Previous studies have primarily focused on the dilated LV as a predictor of poor outcomes in cardiac diseases.<sup>21,22</sup> However, the relationship between small LV size and cardiovascular risks remains elusive, especially in the general population. Our study contributes to this knowledge gap by demonstrating that small LV size, as assessed by CMR, is associated with increased risk of cardiovascular outcomes. Notably, we found that both small and large LV sizes exhibited an elevated risk compared with normal LV, suggesting a U-shaped relationship between LV size and cardiovascular prognosis. This observation aligns with previous



**Figure 4. Associations between LV size and MACE in subgroups.**

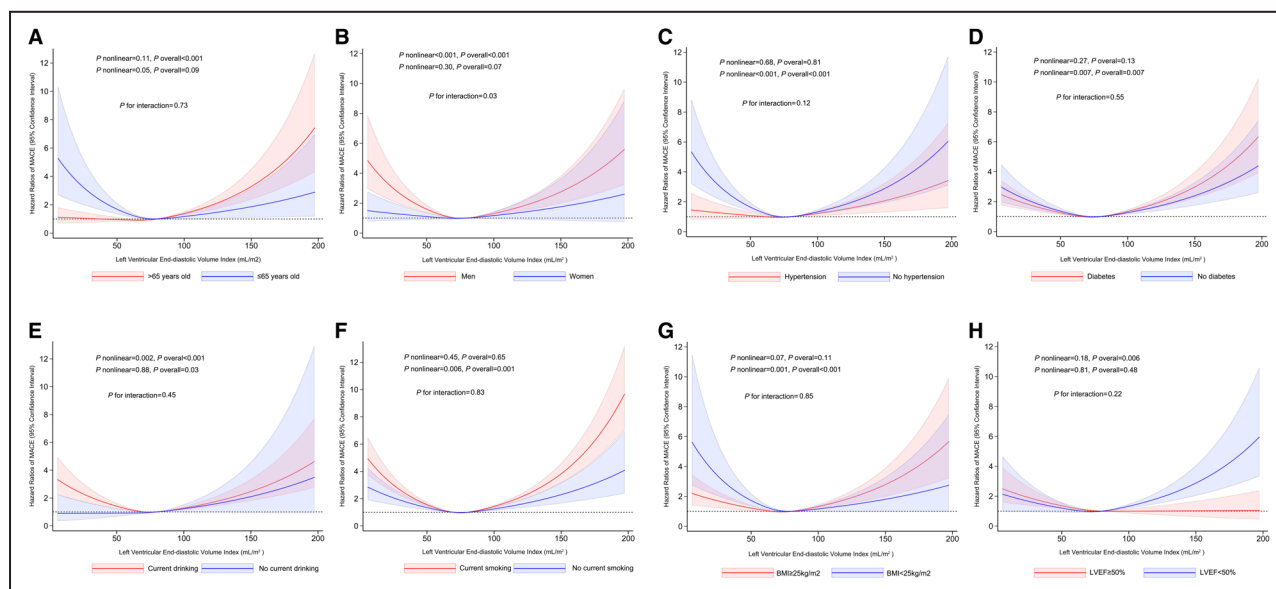
Model adjusted for age, sex, race education, drinking status, smoking status, history of hypertension, diabetes, atrial fibrillation, Townsend deprivation index, physical activity, systolic blood pressure, diastolic blood pressure, body mass index, total cholesterol, hemoglobin concentration, white blood cell count, albumin, hemoglobin A1c, C-reactive protein, use of cardiovascular medications, and left ventricular EF. EF indicates ejection fraction; HR, hazard ratio; LV, left ventricle; and MACE, major adverse cardiovascular events.

studies that reported an increased risk of atrial fibrillation and adverse outcomes in patients with small LVs after transcatheter aortic valve replacement, albeit in specific patient populations.<sup>9,10</sup> However, our study broadens these insights to encompass the general population, thereby underscoring the clinical relevance of LV size across diverse clinical scenarios.

## Potential Mechanisms

The underlying mechanisms linking small LV to adverse cardiovascular outcomes remain to be fully elucidated.

Several reasons may explain this relationship. First, small LV size may reflect reduced cardiac reserve and suboptimal cardiac output, predisposing individuals to ischemia and subsequent cardiovascular events.<sup>23</sup> Second, the emergence of the small LV phenotype was prompted by clinical observations in individuals afflicted with chronic fatigue syndrome, exhibiting diminished LV diameters and stroke volumes.<sup>24,25</sup> This association appears to be primarily mediated by compromised oxygen delivery, disrupted energy metabolism, and impaired hemodynamic control, all of which contribute significantly to the link between circulatory



**Figure 5. Adjusted associations of measures of left ventricular EDVi and MACE in subgroups.**

The reference values (HR: 1) were set at the median value of each subgroup. The solid lines represented the adjusted HRs of left ventricular EDVi across the whole range and the shaded areas represented the corresponding 95% CIs. **A**, Age >65 years vs ≤65. U-shaped relationship between left ventricular EDVi and MACE risk in both age subgroups ( $P$  for interaction=0.73). **B**, Men vs women. Elevated risk at left ventricular EDVi extremes in men, with attenuated association in women ( $P$  for interaction=0.03). **C**, Hypertension vs no hypertension. Persistent U-shaped risk pattern across hypertension status subgroups ( $P$  for interaction=0.12). **D**, Diabetes vs no diabetes. U-shaped relationship between left ventricular EDVi and MACE risk in participants with or without diabetes ( $P$  for interaction=0.55). **E**, Current drinking vs no current drinking. Robust U-shaped relationship in drinkers and nondrinkers ( $P$  for interaction=0.45). **F**, Current smoking vs no current smoking. Risk pattern maintained in smokers and nonsmokers ( $P$  for interaction=0.83). **G**, BMI ≥25 vs <25 kg/m<sup>2</sup>. Association independent of BMI stratification ( $P$  for interaction=0.85). **H**, Left ventricular EF ≥50% vs <50%. U-shaped risk observed regardless of left ventricular EF thresholds ( $P$  for interaction=0.22). BMI indicates body mass index; EDVi, end-diastolic volume index; EF, ejection fraction; HR, hazard ratio; LV, left ventricle; and MACE, major adverse cardiovascular events.

volume deficiency and the severity of fatigue symptom.<sup>23</sup> Third, genetic factors implicated in cardiac chamber morphogenesis may contribute significantly to the heightened cardiovascular risk profile observed in individuals with small LV. A recent study has revealed that genetic variants predictive of low birth weight are associated with reduced chamber volumes, including smaller LV size and diminished stroke volume, thereby conferring a heightened vulnerability to CVD.<sup>26</sup> Lastly, a diminutive heart necessitates intensified contractile efforts to sustain adequate cardiac output, thereby imposing an exaggerated metabolic load on the myocardium, potentially predisposing to an adverse clinical outcome.<sup>27</sup>

Our analysis revealed a pronounced interaction between sex and LV size, significantly affecting the risk of MACE, coronary heart disease, ischemic stroke, and death from any cause. Consequently, heightened vigilance is warranted regarding the elevated risk of adverse outcomes associated with small LV size in men. A plausible underlying mechanism may lie in the protective effects of estrogen in women, exemplified by estrogen's attenuation of LV and cardiomyocyte hypertrophy via an estrogen receptor-mediated pathway

that enhances calcineurin degradation.<sup>28</sup> Furthermore, biomarker profiles exhibit notable disparities between sexes. Specifically, in the general population, women exhibit higher levels of cardiac stretch markers (natriuretic peptides) and fibrosis markers (galectin-3), whereas men display elevated levels of cardiac injury markers (cardiac troponins) and inflammation markers (sST2).<sup>29</sup> Notably, the Y chromosome harbors genes implicated in macrophage activation, influencing innate immunity.<sup>30,31</sup> Systemic inflammation, characterized by macrophage activation and elevated circulating cytokines, contributes to atherosclerosis development.<sup>32</sup> Hence, in men, the mere inheritance of the Y chromosome may confer a genetic predisposition to developing CVDs.

## Clinical Implications

Our findings have important clinical implications. By identifying small LV size as an independent risk factor for adverse cardiovascular outcomes, health care providers can now incorporate this information into their risk assessment strategies. Early identification of individuals with small LV size may enable targeted

interventions aimed at preventing or delaying the onset of CVDs. The concept of a small LV, emerging as a novel structural risk factor, holds particular significance in predicting adverse outcomes and assessing risk profiles, particularly among male individuals. Furthermore, our findings highlight the need for continued investigation into the underlying mechanisms of this relationship, which may lead to the development of novel therapeutic strategies.

## Strengths and Limitations

The strengths of our study lie in its large sample size, comprehensive assessment of LV size using CMR, and rigorous statistical analysis. The use of CMR, the gold standard for assessing cardiac structure and function, allowed for a more accurate assessment of LV size compared with echocardiography. Additionally, our findings were robust, persisting after adjustment for multiple confounding variables. Extensive sensitivity analyses underscored the robust consistency and reliability of the associations discovered between LV size and cardiovascular outcomes, highlighting the paramount importance of small LV in the appraisal and management of cardiovascular risks. These findings have important clinical implications, particularly in emphasizing the need to consider small LV as a prognostic marker for CVD.

Several limitations should be acknowledged. First, the observational nature of our study precludes causal inference. Second, Although CMR was used as the primary imaging modality, potential differences in LV size measurements across different imaging techniques were not explored, which could be a limitation. Additionally, our reliance solely on CMR may have introduced biases related to the specific characteristics of this imaging method. Third, our definition of small LV size was based on normative reference ranges derived from previous studies, which may not be universally applicable. Finally, our study focused on a subset of participants from the UK Biobank, potentially limiting the generalizability of our findings to other populations.

Future research directions could include studies to elucidate the underlying mechanisms of small LV size and its association with CVD, as well as potential interventions to mitigate the risk of adverse cardiovascular events. Such studies would further strengthen the clinical utility of our findings and contribute to the understanding and management of CVD.

## CONCLUSIONS

In the general population, a small LV size emerged as an independent predictor of heightened vulnerability to MACE, coronary heart disease, ischemic stroke, death from any cause, and noncardiovascular death. These observations held true across sex-stratified analyses

and remained robust even after adjustment for conventional cardiovascular risk factors. Thus, recognizing small LV as a pivotal anatomical risk marker may be imperative for refining risk assessment and guiding management strategies in the general populace.

## ARTICLE INFORMATION

Received December 6, 2024; accepted June 3, 2025.

### Affiliations

Department of Guangdong Cardiovascular Institute, Guangdong Provincial People's Hospital (Guangdong Academy of Medical Sciences), Southern Medical University, Guangzhou, China (Y-J.C., Z-L.L., H-Q.W., W-D.L.); Department of Cardiology, the First Affiliated Hospital, Sun Yat-Sen University, Guangzhou, China (L-J.L., Y-D.M., X-M.C., Z-X.W., S-H.W., L-C.W., Y.W.); NHC Key Laboratory of Assisted Circulation, Sun Yat-Sen University, Guangzhou, China (L-J.L., Y-D.M., X-M.C., Z-X.W., S-H.W., L-C.W., Y.W.); The First Clinical Medical College, Guangdong Medical University, Zhanjiang, China (Y-J.L.); and Guangdong Provincial Geriatrics Institute, Guangdong Provincial People's Hospital (Guangdong Academy of Medical Sciences), Southern Medical University, Guangzhou, Guangdong, China (Y-M.X.).

### Acknowledgments

We sincerely thank all the participants of the UK Biobank.

### Sources of Funding

The study was financially supported by the grants from National Natural Science Foundation of China (82270333, 81970280, 81370285 and 81970206), Guangdong Basic and Applied Basic Research Foundation (2024A151013067; 2022A151012358), Natural Science Foundation of Guangdong Province (2022A151012358 and 2019A151010269), Guangdong Provincial Science and Technology Planning Project (Grant No. 2023B110009), the Science and Technology Program of Guangzhou, China (2024B03J1344), High-level Talents Introduction Plan of Guangdong Provincial People's Hospital (KY012023007), Guangzhou Science and Technology Key Project (2023B03J0436), and project of China International Medical Foundation (Z-2016-23-2101).

### Disclosures

None.

### Supplemental Material

Tables S1–S7.

Figures S1–S2.

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